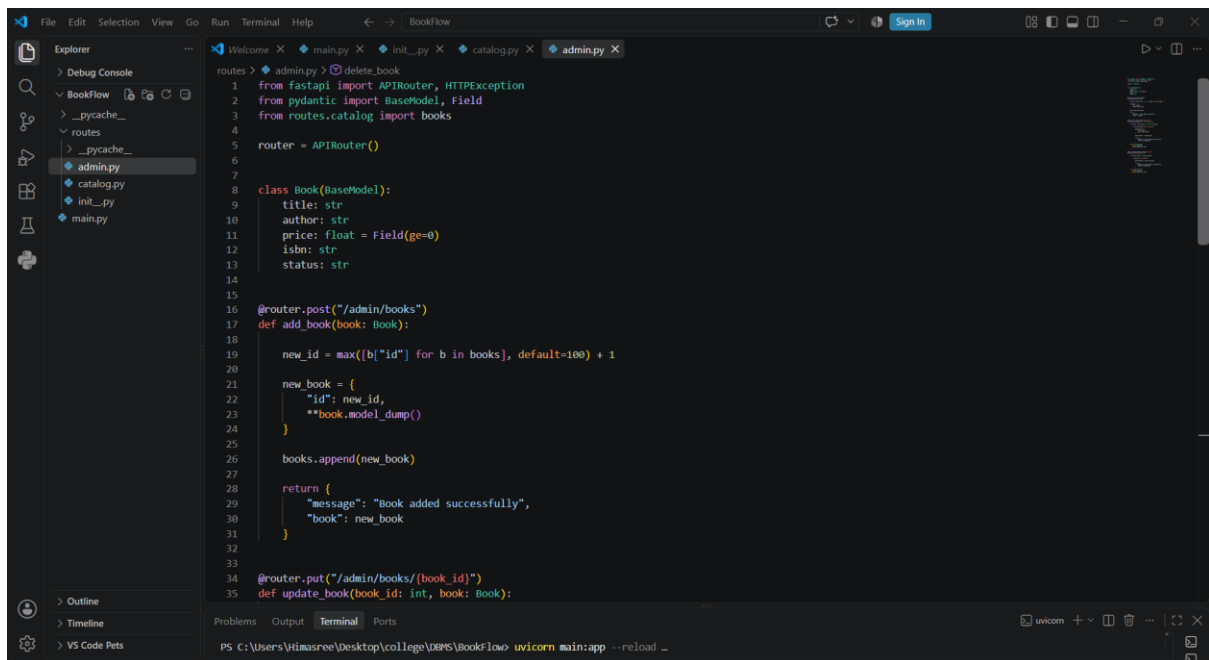


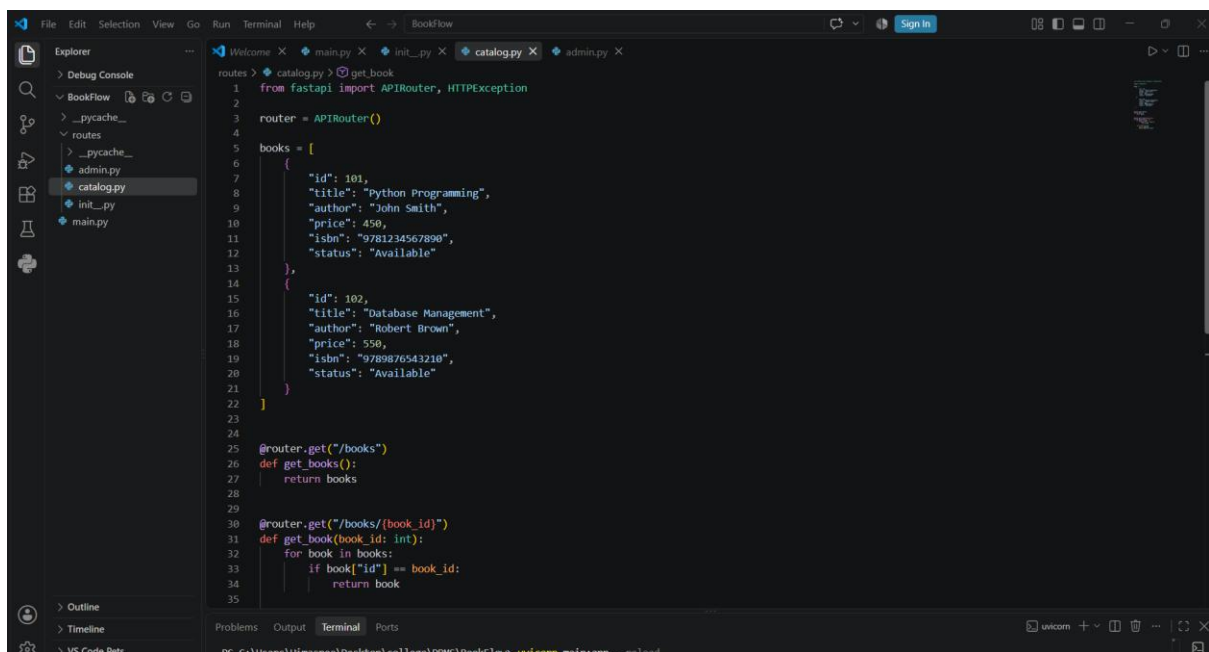
Prcatical-6

Step 1 — Create the project folder



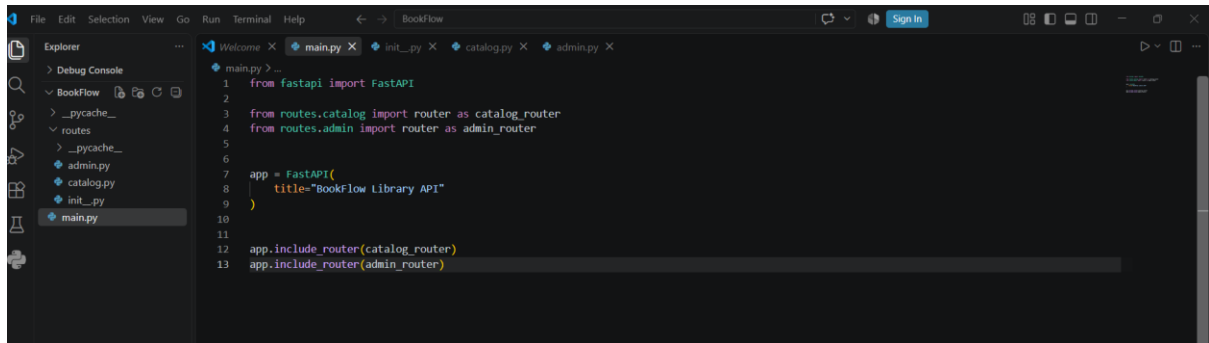
```
1 from fastapi import APIRouter, HTTPException
2 from pydantic import BaseModel, Field
3 from routes.catalog import books
4
5 router = APIRouter()
6
7
8 class Book(BaseModel):
9     title: str
10    author: str
11    price: float = Field(ge=0)
12    isbn: str
13    status: str
14
15
16 @router.post("/admin/books")
17 def add_book(book: Book):
18
19     new_id = max([b["id"] for b in books], default=100) + 1
20
21     new_book = {
22         "id": new_id,
23         **book.model_dump()
24     }
25
26     books.append(new_book)
27
28     return {
29         "message": "Book added successfully",
30         "book": new_book
31     }
32
33
34 @router.put("/admin/books/{book_id}")
35 def update_book(book_id: int, book: Book):
```

Step 2 — Open BookFlow in VS Code



```
1 from fastapi import APIRouter, HTTPException
2
3 router = APIRouter()
4
5 books = [
6     {
7         "id": 101,
8         "title": "Python Programming",
9         "author": "John Smith",
10        "price": 450,
11        "isbn": "9781234567890",
12        "status": "Available"
13    },
14    {
15        "id": 102,
16        "title": "Database Management",
17        "author": "Robert Brown",
18        "price": 550,
19        "isbn": "9789876543210",
20        "status": "Available"
21    }
22 ]
23
24
25 @router.get("/books")
26 def get_books():
27     return books
28
29
30 @router.get("/books/{book_id}")
31 def get_book(book_id: int):
32     for book in books:
33         if book["id"] == book_id:
34             return book
35
```

Step 3 — Create main.py

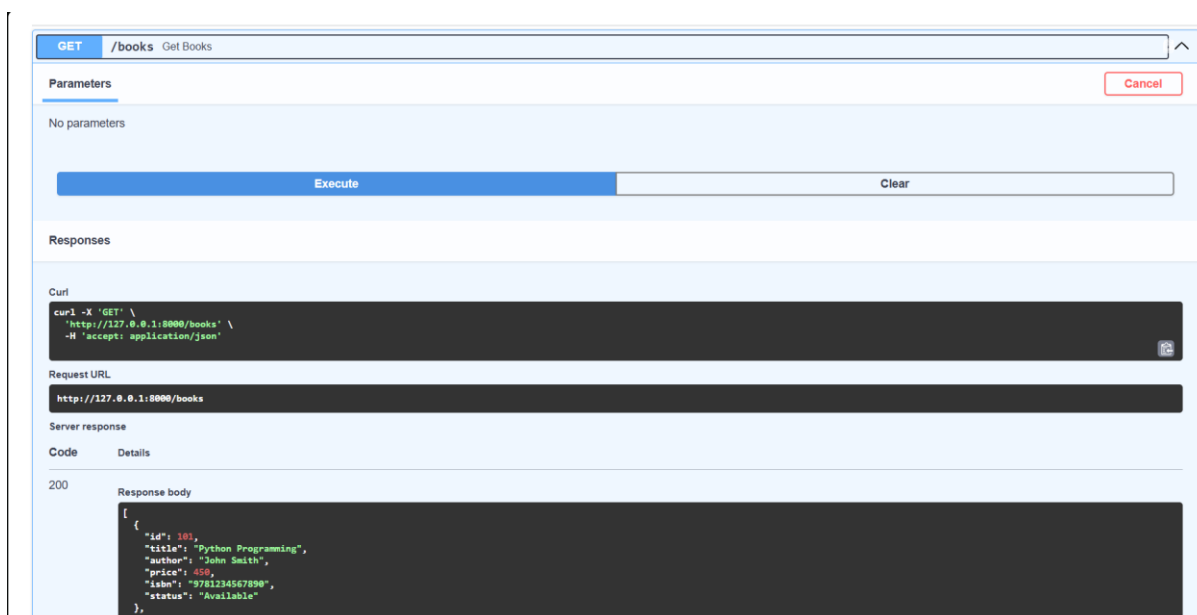


```
1 from fastapi import FastAPI
2
3 from routes.catalog import router as catalog_router
4 from routes.admin import router as admin_router
5
6
7 app = FastAPI(
8     title="BookFlow Library API"
9 )
10
11
12 app.include_router(catalog_router)
13 app.include_router(admin_router)
```

Step 5 — Open Swagger



Step 6 — Test GET all books



Server response

Code	Details
200	Response body <pre>[{ "id": 101, "title": "Python Programming", "author": "John Smith", "price": 450, "isbn": "9781234567890", "status": "Available" }, { "id": 102, "title": "Database Management", "author": "Robert Brown", "price": 550, "isbn": "9789876543210", "status": "Available" }]</pre> Response headers <pre>content-length: 240 content-type: application/json date: Wed, 16 Sep 2026 09:30:54 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls Accept header. Example Value Schema <pre>"string"</pre>	No links

Step 7 — Test GET one book

Name: book_id * required

Integer (path)

[Execute](#) [Clear](#)

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/101' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/101
```

Server response

Code	Details
200	Response body <pre>{ "id": 101, "title": "Python Programming", "author": "John Smith", "price": 450, "isbn": "9781234567890", "status": "Available" }</pre> Response headers <pre>content-length: 117 content-type: application/json date: Wed, 16 Sep 2026 09:32:42 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls Accept header. Example Value Schema <pre>"string"</pre>	No links
422	Validation Error Media type: application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0] }] }</pre>	No links

Step 8— Test POST

POST

/admin/books

Add Book

Parameters

Cancel

Reset

No parameters

Request body

required

application/json

Edit Value

Schema

{

"title": "Operating Systems",

"author": "James Anderson",

"price": 600,

"isbn": "9781111111111",

"status": "Available"

}

Execute

Clear

Responses

Curl

curl -X 'POST' \

'http://127.0.0.1:8000/admin/books' \

-H 'accept: application/json' \

-H 'Content-Type: application/json' \

-d '{

"title": "Operating Systems",

"author": "James Anderson",

"price": 600,

"isbn": "9781111111111",

"status": "Available"

}'

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code

Details

200

Response body

{

"message": "Book added successfully",

"book": {

"id": 103,

"title": "Operating Systems",

"author": "James Anderson",

"price": 600,

"isbn": "9781111111111",

"status": "Available"

}

}

Response headers

content-length: 167

content-type: application/json

date: Wed, 16 Sep 2026 09:34:24 GMT

server: uvicorn

Responses		
Code	Description	Links
200	Successful Response	No links
Media type		
application/json		
Controls Accept header.		
Example Value		
Schema		
"string"		
422	Validation Error	No links
Media type		
application/json		
Example Value		
Schema		
{		
"detail": [		
{		
"loc": [		
"string",		
0		
},		
"msg": "string",		
"type": "string",		
"input": "string",		
"ctx": {}		
}		
]		
}		

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code Details

200

Response body

```
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 450,
  "isbn": "9781234567890",
  "status": "Available"
},
{
  "id": 102,
  "title": "Database Management",
  "author": "Robert Brown",
  "price": 550,
  "isbn": "9789876543210",
  "status": "Available"
},
{
  "id": 103,
  "title": "Operating Systems",
  "author": "James Anderson",
  "price": 600,
  "isbn": "9781111111111",
  "status": "Available"
}
}
```

Response headers

Download

Step 9 — Test PUT

PUT /admin/books/{book_id} Update Book

Parameters

Cancel Reset

Name	Description
book_id required	
integer	103
(path)	

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Advanced Operating Systems",
  "author": "James Anderson",
  "price": 650,
  "isbn": "9781111111111",
  "status": "Available"
}
```

Execute Clear

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/admin/books/103' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Advanced Operating Systems",
    "author": "James Anderson",
    "price": 650,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books/103

Server response

Code Details

200

Response body

```
{
  "message": "Book updated successfully",
  "book": {
    "id": 103,
    "title": "Advanced Operating Systems",
    "author": "James Anderson",
    "price": 650,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```

Response headers

```
content-length: 178
content-type: application/json
date: Wed, 16 Sep 2026 09:36:03 GMT
server: uvicorn
```

Responses

Code	Description
------	-------------

Links

Responses		
Code	Description	Links
200	Successful Response	No links
Media type application/json		
Controls Accept header.		
Example Value Schema		
"string"		
422	Validation Error	No links
Media type application/json		
Example Value Schema		
{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }		

Step 10 — Test DELETE

DELETE /admin/books/{book_id} Delete Book		
Parameters		
Name	Description	
book_id * required integer (path)	103	
Execute		Clear
Responses		
Curl curl -X 'DELETE' \ 'http://127.0.0.1:8000/admin/books/103' \ -H 'accept: application/json'		
Request URL http://127.0.0.1:8000/admin/books/103		
Server response		
Code	Details	
200	Response body	
Server response		
Code	Details	
200	Response body	
{ "message": "Book deleted successfully", "book": { "id": 103, "title": "Advanced Operating Systems", "author": "James Anderson", "price": 650, "isbn": "9781111111111", "status": "Available" } }		
Response headers content-length: 178 content-type: application/json date: Wed, 16 Sep 2026 09:36:45 GMT server: uvicorn		
Responses		
Code	Description	Links
200	Successful Response	No link
Media type application/json		
Controls Accept header.		
Example Value Schema		
"string"		

Step 11 — Verify deletion

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

200

Response body

```
[
  {
    "id": 101,
    "title": "Python Programming",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  },
  {
    "id": 102,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 550,
    "isbn": "9789876543210",
    "status": "Available"
  }
]
```

Code

Details

200

Response body

```
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 450,
  "isbn": "9781234567890",
  "status": "Available"
},
{
  "id": 102,
  "title": "Database Management",
  "author": "Robert Brown",
  "price": 550,
  "isbn": "9789876543210",
  "status": "Available"
}
]
```

Response headers

```
content-length: 240
content-type: application/json
date: Wed, 16 Sep 2026 09:37:17 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header.

Example Value | Schema